

NorthStar GAP Tool

Formula Reference • May 2026

Authoritative reference for every formula used in metric cards and visualizations across all modules.

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1. Life Insurance Module

1.1 Core Gap Calculator

Years to Retirement

$$\text{yearsToRetirement} = \text{retirementAge} - \text{currentAge}$$

The number of years from the client's current age to their target retirement age. All income and coverage projections are computed over this window.

Planning Horizon

Annual Client Need

$$\text{annualClientNeed} = \text{annualIncome} \times \text{incomeReplacementRatio}$$

Scales the client's gross annual income by the desired income-replacement ratio to produce the yearly dollar amount the survivor family would need.

Metric Card

Annual Replacement Need

$$\text{annualReplacementNeed} = \max(0, \text{annualClientNeed} - \text{spouseAnnualIncome})$$

Offsets the annual client need by the surviving spouse's independent income to prevent double-counting.

Metric Card

Future Income Lost (PV Mode)

$$\text{futureIncomeLost} = \frac{\sum_{\text{year}=1}^{\text{replacementYears}} \text{annualReplacementNeed} \times (1 + \text{incomeGrowthRate})^{\text{year}}}{(1 + \text{discountRate})^{\text{year}}}$$

Each future year's replacement need is grown by income growth rate and discounted back to today's dollars. Summing yields a lump-sum equivalent.

PV Assumption

Metric Card

Future Income Lost (Simple Mode)

$$\text{futureIncomeLost} = \text{annualReplacementNeed} \times \text{replacementYears}$$

When present-value discounting is disabled, multiplies constant annual replacement need by the number of replacement years.

Simple Mode

Base Protection Need

$$\text{baseProtectionNeed} = \text{futureIncomeLost} + \text{debtsTotal} + \text{educationGoal} + \text{finalExpenses}$$

Adds outstanding debts, education funding goals, and final expenses to the income-replacement component.

Total Death Benefit Need

Available Resources & Gap

$$\begin{aligned} \text{availableResources} &= \text{existingCoverageTotal} + \text{liquidAssetOffset} \\ \text{remainingGap} &= \max(0, \text{baseProtectionNeed} - \text{availableResources}) \\ \text{coverageOffsetPercentage} &= \text{availableResources} \div \text{baseProtectionNeed} \end{aligned}$$

Available resources include existing coverage plus optional liquid assets. The gap is the shortfall; offset percentage shows the funded fraction.

Headline Metrics

Annuity Payment (Income Yield)

```
annuityPayment = PV × r / (1 - (1+r)^(-n))  
If r = 0: annuityPayment = PV / n
```

Converts a lump-sum death benefit into the equivalent level annual income stream over the coverage period.

Income Yield

Survivor Gap & Cumulative Metrics

```
survivorGap[year] = max(0, projectedIncome[year] - gliCovered[year] - privateCovered[year])  
cumulativeSurvivorGap = Σ survivorGap[year]  
lifetimeIncomeUncoveredPercentage = cumulativeSurvivorGap ÷ projectedIncomeToRetirement
```

Calculates annual and cumulative uncovered income, plus the percentage of total lifetime income left unprotected.

Chart & Cards

1.2 Income Gap Scenarios

Funding Base & Net Need

```
coverageFundingBase = groupLifeCoverage + privateLifeCoverage + nonQualifiedAssets
annualNetIncomeNeed = max(0, annualIncome × incomeReplacementRatio - spouseAnnualIncome)
```

The investable pool available to the survivor. Life insurance death benefits are tax-free, so no gross-to-net conversion is applied.

Module 1 — Safe Withdrawal Rate

```
annualSafeWD = coverageFundingBase × r / (1 - (1+r)^(-years))
If r = 0: annualSafeWD = fundingBase / years
m1TotalReplaced = annualSafeWD × yearsToRetirement
m1SurvivorGap = max(0, projectedNetIncomeTotal - m1TotalReplaced)
```

Level annual income drawn from funding base over the full horizon, assuming full depletion at retirement.

Module 2 — Max Withdrawal (Aggressive)

```
// Annual cycle:
balance = balance × (1 + maxROI)
covered = min(balance, projectedNetIncome)
balance = max(0, balance - covered)
```

Fund grows at ROI, then survivor draws full need. Continues until exhaustion. Fully covered years are green; shortfalls are red.

Present Value of Level Gap

```
annualGap = survivorGap ÷ yearsToRetirement
DB_Needed = annualGap × (1 - (1+roi)^(-years)) / roi
If roi = 0: DB_Needed = survivorGap
```

Converts cumulative survivor gap to equivalent level annual gap, then discounts to present-value lump sum needed today.

2. Disability Insurance Module

2.1 Core Gap Calculator

LTD Monthly Gross & Net

```
ltdMonthlyGross = min(annualIncome × coveragePercent ÷ 12, monthlyCap)
ltdNetMonthly = ltdMonthlyGross × 0.70 (if taxable)
ltdNetMonthly = ltdMonthlyGross (if non-taxable)
```

Applies coverage percentage, caps benefit, and applies simplified 30% tax haircut if applicable.

Income Loss & Projections

```
totalNetMonthly = ltdNetMonthly + privateDiMonthly
incomeLossNet = (annualIncome × 0.70 ÷ 12) - totalNetMonthly
annualIncomeNetAtAge[year] = annualIncome × (1+growthRate)^year × 0.70
```

Immediate monthly shortfall vs combined benefits. Projects after-tax income forward for coverage comparison.

Annual Gap & Cumulative Metrics

```
annualGap[year] = max(0, netIncome - ltdAnnualBenefit - idIAnnualBenefit)
totalGap = Σ annualGap[year]
averageCoverageRate = totalCoverage ÷ totalProjectedIncome
```

Unprotected net income at each age. Cumulative gap shows maximum lifetime exposure. Average rate shows coverage efficiency.

2.2 Break-Even / Premium vs. Self-Insured

Investment Balance & Benefits

```
investmentBalance[m] = (balance[m-1] + premium) × (1 + monthlyRate)
benefitsReceived = monthlyBenefit × min(monthsWithoutIncome, 80)
```

Simulates investing premiums in a side fund. Benefits capped at 80 months per lookup table boundary.

Break-Even Point

```
breakEvenMonths = log((premium + benefits × rate) / premium) / log(1 + rate)
If rate = 0: breakEvenMonths = benefits ÷ premium
```

Months needed to self-insure before accumulated fund matches insurance benefit. Mirrors Excel NPER function.

2.3 Premium vs. Self-Insured — Retirement Projection

Future Value & Coverage Months

```
FV = monthlyPremium × ((1 + r/12)^n - 1) / (r/12)
monthsOfCoverage = FV ÷ monthlyBenefit
yearsOfCoverage = monthsOfCoverage ÷ 12
```

Standard future-value of an ordinary annuity. Shows equivalent disability months the invested premiums could fund at retirement.

2.4 Savings Bridge (Elimination Period)

Elimination Period & Runway

```
eliminationMonths = days ÷ 30.4375
savingsNeeded = monthlyExpenses × eliminationMonths
savingsRunway = liquidSavings ÷ monthlyExpenses
postBridgeRunway = savingsRemaining ÷ ongoingMonthlyGap
```

Models liquid savings depletion during waiting period. Calculates post-LTD monthly gap and remaining runway.

2.5 SSDI Benefit Estimator

PIA Calculation (2025 Bend Points)

```
AIME ≈ careerAvgAnnual ÷ 12
Tier1 = 0.90 × min(AIME, 1226)
Tier2 = 0.32 × max(0, min(AIME, 7391) - 1226)
Tier3 = 0.15 × max(0, AIME - 7391)
PIA = round(Tier1 + Tier2 + Tier3, $0.10)
Monthly SSDI = min(PIA, 3822)
```

Implements SSA formula using 2025 brackets. Applies statutory maximum cap of \$3,822/mo.

Replacement Rate & Waiting Period

```
replacementRateGross = Monthly SSDI ÷ (annualIncome ÷ 12)
monthlyGap = max(0, monthlyIncome - Monthly SSDI)
waitingLoss = (annualIncome ÷ 12) × 5
```

Calculates income gap not covered by SSDI and income lost during mandatory 5-month waiting period.

2.6 LTD Benefit Taxation Calculator

Tax Computation & Replacement Rates

```
federalTax = piecewise bracket tax on (grossBenefit - stdDeduction)
stateTax = taxableBenefit × stateRate
netAnnualBenefit = annualGrossBenefit - totalTax
netReplacementRate = netAnnualBenefit ÷ preTaxAnnualIncome
```

Calculates actual after-tax take-home pay during disability. Compares net LTD to net pre-disability income for true replacement rate.

2.7 Job Comparison Module

Job A vs Job B Protection

```
// Job A (No IDI):  
groupCovered_A = min(groupLTD_A, salary_A)  
gap_A = salary_A - groupCovered_A  
  
// Job B (With IDI):  
effectiveSalary_B = salary_A - premium_B  
groupCovered_B = min(groupLTD_B, effectiveSalary_B)  
idiCovered_B = min(IDI_B, max(0, effectiveSalary_B - groupCovered_B - premium_B))  
gap_B = max(0, effectiveSalary_B - groupCovered_B - idiCovered_B)  
  
advantage = (groupCovered_B + idiCovered_B) - groupCovered_A
```

Models real take-home impact after IDI premium. Shows additional annual disability income available in Job B vs A.

3. Liability Module

Wage Garnishment Risk

```
incomeRisk = Σ annualIncome × (1+growth)^year × garnishmentRate
householdWageRisk = primaryIncomeRisk + spouseIncomeRisk
```

Cumulative income at risk of garnishment over remaining career using default 25% maximum legal limit.

Assets at Risk & Total Liability

```
nonQualifiedRisk = max(inputs.nonQualified, investments + savings + business)
totalHouseholdLiability = householdWageRisk + nonQualifiedRisk
totalCoverage = autoLiability + umbrella
```

Excludes qualified retirement accounts. Sums wage exposure and non-protected assets to find total exposure.

Gap & Recommendation Logic

```
liabilityGap = max(0, totalLiability - totalCoverage)
recommendedUmbrella = max(netWorthAtRisk, income×5, 1000000)
shortfall = max(0, recommendedUmbrella - currentUmbrella)
```

Gap shows unprotected exposure. Recommended umbrella uses multi-factor approach: net worth, 5x income, or \$1M minimum.

4. Unemployment Module

Cash Flow & Reserves

```
monthlyCashFlow = monthlyIncome - burnRate
minReserve = burnRate × 3
optimalReserve = burnRate × 6
dangerThreshold = burnRate × 1.5
runwayRaw = emergencySavings ÷ burnRate
```

Baseline surplus/deficit. Standard benchmarks for emergency fund adequacy. Savings-only runway calculation.

Offsets & Effective Runway

```
monthlyOffset = severance + unemploymentBenefit
netBurn = burnRate - monthlyOffset
totalOffsetPool = (severance × duration) + (benefit × duration)
effectiveRunway = (savings + totalOffsetPool) ÷ burnRate
```

Combines severance and state benefits. Calculates theoretical upper-bound runway when all resources are pooled.

Month-by-Month Simulation

```
required[m] = max(0, burnRate - offsetIncome[m])
covered[m] = min(reserveBalance, required[m])
shortfall[m] = max(0, required[m] - reserveBalance)
reserveBalance[m] = max(0, reserveBalance[m-1] - required[m])
```

Granular monthly tracking of offset income, savings drawdown, and depletion timeline.